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Student Number



NEWINGTON COLLEGE

2014
HSC Assessment 1
Year 12 Mathematics

General Instructions:

- Date of task – Tuesday 18th November (Period 6)
- Working time – 45 mins
- Weighting - 15%
- Board-approved calculators may be used.
- Start each question in a new booklet.
- Show all relevant mathematical reasoning and/or calculations.

Total marks – 30

Outcomes to be assessed:

- P6** Relates the derivative of a function to the slope of its graph.
- P7** Determines the derivative of a function through routine application of the rule of differentiation.
- P8** Understands and uses the language and notation of calculus.
- H6** Uses the derivative to determine the features of the graph of a function.
- H7** Uses the features of a graph to deduce information about the derivative.

Section A (4 marks)

Multiple choice, answer on the multiple choice answer sheet provided.

1) What is the equation of the directrix of the parabola $y^2 = -12x$?

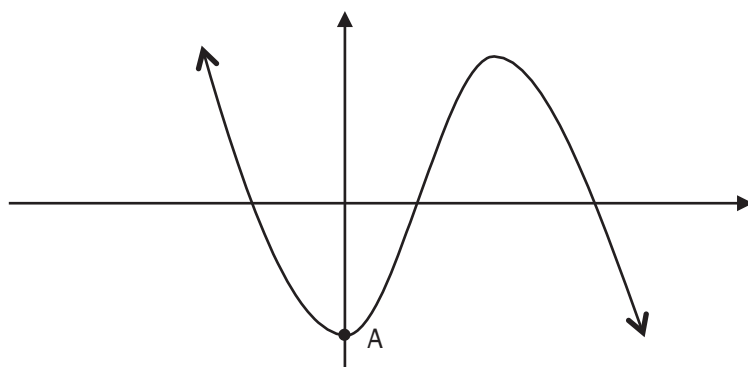
(A) $x = -3$

(C) $y = -3$

(B) $x = 3$

(D) $y = 3$

2) At the point A on the function shown below which of the following is true?



(A) $f'(x) < 0, f''(x) < 0$

(C) $f'(x) = 0, f''(x) = 0$

(B) $f'(x) = 0, f''(x) > 0$

(D) $f'(x) = 0, f''(x) < 0$

3) Which of the following is the second derivative of $\frac{1}{\sqrt{x}}$?

(A) $-\frac{1}{2\sqrt{x^3}}$

(C) $\frac{3}{4}\sqrt{x^5}$

(B) $-\frac{3}{4\sqrt{x^5}}$

(D) $\frac{3}{4\sqrt{x^5}}$

4) For what values of x is the curve $f(x) = 2x^3 + x^2$ concave down?

(A) $x < -\frac{1}{6}$

(C) $x < -6$

(B) $x > -\frac{1}{6}$

(D) $x > -6$

Section B (13 marks)

Answer in a new booklet.

- 1) If the quadratic equation $2x^2 - 3x + 4 = 0$ has the roots α and β , find
- (i) $\alpha\beta$ (1 mark)
 - (ii) $\alpha + \beta$ (1 mark)
 - (iii) $4\alpha^2\beta + 4\alpha\beta^2$ (2 marks)
- 2) Find A and B given that $x^2 + x \equiv A(x^2 - 3x) + Bx$. (2 marks)
- 3) Consider the quadratic equation $(k+3)x^2 + 4x + k = 0$.
- (i) Show that the discriminant is $16 - 12k - 4k^2$ (1 mark)
 - (ii) Find the value(s) of k for which the equation $(k+3)x^2 + 4x + k = 0$ has real roots. (2 marks)
- 4) A and B are the points $(1, 4)$ and $(-3, 2)$ respectively. The point $P(x, y)$ moves such that PA is perpendicular to PB .
- (i) Show that the equation of the locus of P is the circle $x^2 + 2x + y^2 - 6y + 5 = 0$ (2 marks)
 - (ii) Find the radius and centre of this circle. (2 marks)

(End of section B)

Section C (13 marks)

Answer in a new booklet.

- 1) Find the first derivative of $f(x) = x(5x - 1)^4$, giving your answer in its factorised form. (2 marks)

- 2) For the parabola $(x + 2)^2 = -6y + 24$, find the
 - (i) coordinates of the vertex (1 mark)
 - (ii) coordinates of the focus (1 mark)

- 3) Consider the curve given by $y = 3x^2 - x^3 + 9x - 2$.
 - (i) Find the co-ordinates of the stationary points and determine their nature. (3 marks)
 - (ii) Determine the co-ordinates of any point(s) of inflexion. (2 marks)
 - (iii) Sketch the graph of $y = 3x^2 - x^3 + 9x - 2$, clearly indicating its stationary points, inflexion point(s) and y-intercept. (2 marks)

- 4) The line $5x + 2y + 14 = 0$ forms a focal chord of the parabola $(y - 3)^2 = -4a(x + 2)$. By using the focus, find a , the focal length of the parabola. (2 marks)

(End of section C)

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Multiple Choice Answer Sheet - Mathematics

Completely fill the response oval representing the most correct answer.

- 1 A ☐ B ☐ C ☐ D ☐
- 2 A ☐ B ☐ C ☐ D ☐
- 3 A ☐ B ☐ C ☐ D ☐
- 4 A ☐ B ☐ C ☐ D ☐

Year 12 Mathematics Assessment Task 1 Solutions and Marking Criteria

Section A

Q.	Solution	Marking criteria
1	$x = 3 \therefore B$	1 mark for correct answer
2	$f'(x) = 0, f''(x) > 0 \therefore B$	1 mark for correct answer
3	$y = x^{-\frac{1}{2}}$ $y' = -\frac{1}{2}x^{-\frac{3}{2}}$ $y'' = \frac{3}{4}x^{-\frac{5}{2}}$ $= \frac{3}{4\sqrt{x^5}} \therefore D$	1 mark for correct answer
4	$f'(x) = 6x^2 + 2x$ $f''(x) = 12x + 2$ $12x + 2 > 0$ $12x > -2$ $x > -\frac{1}{6} \therefore A$	1 mark for correct answer

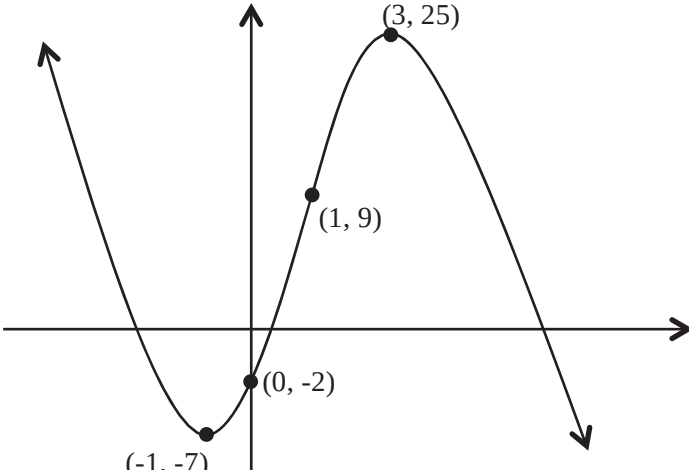
Section B

Q.	Solution	Marking criteria
1 (i)	$\alpha\beta = \frac{c}{a} = \frac{4}{2}$ $= 2$	1 mark for correct answer
(ii)	$\alpha + \beta = -\frac{b}{a} = -\frac{-3}{2}$ $= \frac{3}{2}$	1 mark for correct answer
(iii)	$4\alpha^2\beta + 4\alpha\beta^2 = 4\alpha\beta(\alpha + \beta)$ $= (4 \times 2) \times \frac{3}{2}$ $= 12$	1 mark for correct factorisation 1 mark for correct answer
2	$A(x^2 - 3x) + Bx$ $= Ax^2 - 3Ax + Bx$ $= Ax^2 + (B - 3A)x$ $\therefore A = 1$ $\therefore B - 3A = 1$ $B - 3 = 1$ $B = 4$	1 mark for $A = 1$ 1 mark for $B = 4$

3 (i)	$\Delta = b^2 - 4ac$ $= 4^2 - 4 \times (k+3) \times k$ $= 16 - 4k(k+3)$ $= 16 - 4k^2 - 12k$ $= 16 - 12k - 4k^2$	1 mark for correctly showing the discriminant
(ii)	$\Delta \geq 0$ $16 - 12k - 4k^2 \geq 0$ $4 - 3k - k^2 \geq 0$ $k^2 + 3k - 4 \leq 0$ $(k+4)(k-1) \leq 0$ $\therefore -4 \leq k \leq 1$	1 mark for $16 - 12k - 4k^2 \geq 0$ or equivalent 1 mark of solution follows
4 (i)	$m_{PA} = \frac{y-4}{x-1}, \quad m_{PB} = \frac{y-2}{x+3}$ $m_{PA} \times m_{PB} = -1$ $\frac{y-4}{x-1} \times \frac{y-2}{x+3} = -1$ $(y-4)(y-2) = -(x-1)(x+3)$ $y^2 - 6y + 8 = -(x^2 + 2x - 3)$ $y^2 - 6y + 8 = -x^2 - 2x + 3$ $\therefore x^2 - 2x + y^2 - 6y + 5 = 0$	1 mark for $\frac{y-4}{x-1} \times \frac{y-2}{x+3} = -1$ or equivalent 1 mark for showing the correct equation of the circle
(ii)	$(x+1)^2 + (y-3)^2 = -5 + 1 + 9$ $= 5$ Centre $(-1, 3)$ Radius $\sqrt{5}$ units	1 mark for centre 1 mark for radius

Section C

Q.	Solution	Marking criteria
1	$f'(x) = x \times [4(5x-1)^3 \times 5] + 1 \times (5x-1)^4$ $= 20x(5x-1)^3 + (5x-1)^4$ $= (5x-1)^3(20x+5x-1)$ $= (5x-1)^3(25x-1)$	1 mark for correct application of the product rule and chain rule 1 mark for correct factorisation
2 (i)	$(x+2)^2 = -6y + 24$ $(x+2)^2 = -6(y-4)$ vertex is at $(-2, 4)$	1 mark for correct answer
(ii)	$a = \frac{6}{4} = \frac{3}{2}$ focus is at $\left(-2, \frac{5}{2}\right)$	1 mark if focus follows using $a = \frac{3}{2}$

3 (i)	$y' = 6x - 3x^2 + 9$ $y'' = -6x + 6$ Stationary points when $y' = 0$ $-3x^2 + 6x + 9 = 0$ $3x^2 - 6x - 9 = 0$ $x^2 - 2x - 3 = 0$ $(x - 3)(x + 1) = 0$ $\therefore x = 3 \text{ or } x = -1$ When $x = 3$, $y'' = -6 \times 3 + 6 = -12$ $\therefore$ curve is concave down at $x = 3$ When $x = -1$, $y'' = -6 \times -1 + 6 = 12$ $\therefore$ curve is concave up at $x = -1$ When $x = 3$, $y = 25$ When $x = -1$, $y = -7$ $\therefore$ local minimum turning point at $(-1, -7)$ local maximum turning point at $(3, 25)$	1 mark for correct x values of stationary points 1 mark for correctly determining the nature of the stationary points 1 mark for correctly stating the co-ordinates of the stationary points
(ii)	Let $y'' = 0$ $-6x + 6 = 0$ $-6x = -6$ $x = 1$ Possible point of inflexion at $x = 1$ Testing, let $\varepsilon = 0.1$ At $x + \varepsilon$, $y'' < 0$ $x - \varepsilon$, $y'' > 0$ $\therefore$ point of inflexion exists At $x = 1$, $y = 9$ $\therefore$ point of inflexion at $(1, 9)$	1 mark for correct co-ordinates of point of inflexion 1 mark for checking for concavity change and hence concluding point of inflexion occurs
(iii)		1 mark for correct shape of curve indicating the stationary points 1 mark for labelling inflexion point and y-intercept

4	Vertex $(-2, 3)$ Focus $(x, 3)$ When $y = 3$, $5x + 6 + 14 = 0$ $5x = -40$ $x = -4$ $\therefore$ focus is at $(-4, 3)$ $\therefore$ focal length, $a = 2$ units	1 mark for identifying the y value of the focus 1 mark for finding the focal length
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